F Distribution and ANOVA

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Question 1

0/1 pt 3 19

Fill in the missing values for this ANOVA source table, and use it to calculate the effect size:

	SS	*df*	*MS* (S^2)	*F*
Between	163.725	3		
Subject	554.4	11		
Error				
TOTAL	1543.125			

R^2 =

Question 2

0/1 pt 3 19

Below is the raw data for an experiment. Conduct the appropriate ANOVA hypothesis test with α=0.01, calculating the appropriate bracketed terms, and filling in the source table.

If decimals exceed 3 places, round to 3 places.

Participant	Black	G1	G2	G3	White
A	1	4	1	3	2
B	5	1	4	6	3
C	2	6	6	2	6
D	6	2	5	5	6

To get $[Y]$, square each term, then add them up all up: $[Y]$ = ______

For $[A]$, add up each column, then square those numbers, then divide by the number of participants: $[A]$ = ______

For $[S]$, you follow the same process, but flip your rows and columns: $[S]$ = ______

For $[T]$, add up every datapoint, square that number, and divide by the total number of datapoints.: $[T]$ = ______

Here's some cool formulas:

$$SS_{Bet} = [A] - [T]$$

$$SS_{Sub} = [S] - [T]$$

$$SS_{Tot} = [Y] - [T]$$

Source of Variation	*SS*	*df*	*MS (S^2)*	*F*
Between Groups	______	______	______	______
Subjects	______	______	______	______
Error	______	______	______	
Total	______	______		

F_{crit} = ______

Based upon the results of your ANOVA, do you reject, or fail to reject the null hypothesis?

- Do not reject
- Reject

What is the effect size of this study?

R^2 = ______

Hint

Question 3

0/1 pt 3 19

Below is the raw data for an experiment. Conduct the appropriate ANOVA hypothesis test with α=0.01, calculating the appropriate bracketed terms, and filling in the source table.

If decimals exceed 3 places, round to 3 places.

Participant	Black	G1	G2	G3	White
A	6	5	5	4	2
B	3	4	6	4	3
C	2	4	8	5	2
D	3	4	10	5	2

To get $[Y]$, square each term, then add them up all up: $[Y]$ = ____________

For $[A]$, add up each column, then square those numbers: $[A]$ = ____________

For $[S]$, you follow the same process, but flip your rows and columns: $[S]$ =

____

For $[T]$, add up every datapoint, square that number, and divide by the total number of datapoints.: $[T]$ = ____________

Here's some cool formulas:

$$SS_{Bet} = [A] - [T]$$

$$SS_{Sub} = [S] - [T]$$

$$SS_{Tot} = [Y] - [T]$$

Source of Variation	***SS***	***df***	***MS (S^2)***	***F***
Between Groups	____________	________	____________	____________
Subjects	____________	________	____________	____________
Error	____________	________	____________	
Total	____________	________		

F_{crit} = ______

Based upon the results of your ANOVA, do you reject, or fail to reject the null hypothesis?

- Reject
- Do not reject

What is the effect size?

R^2 = ______

Question 4

0/1 pt 3 19

Below is the raw data for an experiment. Conduct the appropriate ANOVA hypothesis test with α=0.01, calculating the appropriate bracketed terms, and filling in the source table.

If decimals exceed 3 places, round to 3 places.

Participant	Black	G1	G2	G3	White
A	2	4	8	7	8
B	6	5	9	6	8
C	6	5	6	6	7
D	4	4	6	7	7

To get $[Y]$, square each term, then add them up all up: $[Y]$ = ______

For $[A]$, add up each column, then square those numbers: $[A]$ = ______

For $[S]$, you follow the same process, but flip your rows and columns: $[S]$ = ______

For $[T]$, add up every datapoint, square that number, and divide by the total number of datapoints.: $[T]$ = ______

Here's some cool formulas:

$$SS_{Bet} = [A] - [T]$$

$$SS_{Sub} = [S] - [T]$$

$$SS_{Tot} = [Y] - [T]$$

Source of Variation	***SS***	***df***	***MS (S²)***	***F***
Between Groups	______	______	______	______
Subjects	______	______	______	______
Error	______	______	______	
Total	______	______		

F_{crit} = ______

Based upon the results of your ANOVA, do you reject, or fail to reject the null hypothesis?

- Reject
- Do not reject

What is the effect size of this study?

R^2 = ______

Question 5

0/1 pt 3 19

What is the F_{crit} value for a between groups study with 5 groups, having 6 participants per group, and a level of significance (α) of 0.1?

Use software such as Excel if your table is not specific enough: F.INV.RT(α,df_{between},df_{within}).

If using software, round to two decimal places.

F_{crit} = ______

Question 6

0/1 pt 3 19

What is the F_{crit} value with for a **within groups** study with 2 groups, having 9 participants total, and a level of significance (α) of 0.01?

If using software, round to two decimal places.

F_{crit} = ______

Question 7

0/1 pt 3 19

A dean wants to check the consistency of teaching quality among the faculty in one of the departments. A sample of students was obtained from five classes taught by five different faculty, and their common final exam scores were recorded.

Classes	Final Exam Scores									
Class 1	73	77	64	78	79	73	76	73	72	65
Class 2	71	53	64	66	81	54	82	73	62	77
Class 3	74	65	78	68	67	58				
Class 4	75	64	78	70	62	75	67			
Class 5	70	90	62	85	86	71				

Can you conclude that the teaching quality is inconsistent among the courses? Use the level of significance 5%.

Procedure: Select an answer

Assumptions: (select everything that applies)

- Independent samples
- Simple random samples
- At most 20% of the expected frequencies are less than 5
- All expected frequencies are 1 or greater
- Normal populations
- Equal population standard deviations

Question Help: Video

Question 8 0/1 pt 3 19

This problem is partially based on problems 13.1, 13.2, & 13.3 from Lomax & Hahs-Vaughn, 3rd ed.

Complete the following ANOVA summary table for a two-factor fixed-effects ANOVA, where there are three levels of factor A (school) and seven levels of factor B (curriculum design). Each cell includes 15 students.

Source	SS	df	MS	F	p
A	3314.4			3.965	0.02
B	5064.8			2.019	0.063
$A \times B$	6946.2			1.385	0.172
Error					
TOTAL	138217.4				

Hint 1

Hint 2